

Roman Penunuri

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SUMMARY

Fourth-year Electrical Engineering student with verified field experience in safety-critical industrial electrical systems, precision hardware configuration, and compliance documentation. OSHA-30 certified with direct exposure to high-consequence environments requiring meticulous documentation standards and work that must be verified before deployment. Experienced resolving protection system failures through systematic hardware analysis and settings validation. Comfortable operating under rigorous safety requirements, coordinating across multidisciplinary teams, and producing technical deliverables that meet formal review standards. Self-motivated and driven to see technical problems through from start to finish, proactive about seeking input, and comfortable integrating ideas across disciplines.

EDUCATION

University of Houston, Cullen College of Engineering

B.S. Electrical Engineering, Power Systems Concentration | GPA: 3.82

Dean's List: Fall 2024, Spring 2026

Houston, TX

May 2028

Relevant Coursework: Digital Logic Design, Microprocessor Systems, Applied Electromagnetic Waves (Honors), Signals & Systems Analysis, Numerical Methods, Engineering Statistics, Automatic Control Systems (in progress)

SKILLS

Certifications: OSHA-30 General Industry | SEL CBT 101 Protective Relay Fundamentals (4 PDH) | Pursuing ETAP Operator Certification

Proficient: GE Vernova EnerVista (Multilin), MATLAB, C, Falstad, LTSpice, MS Suite

Currently Learning: KiCad (schematic capture and PCB layout), Python, ETAP, PLC Programming

Languages: Fluent in English, Conversational in Spanish

PROJECTS

Water Level Indicator: BJT Sensing Circuit

University of Houston

- Designed and characterized a BJT transistor-based water level sensing circuit; diagnosed LED underperformance by returning to Falstad simulation, probing transistor nodes with an oscilloscope, and iterating between simulation and bench hardware to isolate a saturation failure caused by unexpectedly high distilled water resistivity

555 Timer PCB: KiCad Design Workshop

IEEE UH Chapter

- Completed KiCad schematic capture and PCB layout for a 555 timer astable circuit, progressing from component placement and routing through Gerber file generation as part of an IEEE design workshop

WORK EXPERIENCE

Air Liquide

Pasadena, TX

Plant Operations Electrical Engineering Intern

June 2025 - August 2025

- Diagnosed recurring motor protection failures by systematically analyzing a GE Vernova Multilin 469 relay configuration; executed full settings migration and verification to the 869 platform in EnerVista, confirming parity across all protection elements, eliminating the documented fault source and reducing projected maintenance costs by ~\$10,000
- Produced scope-of-work packages for industrial turnaround operations covering motors, switchgear, and battery systems, incorporating NFPA 70E and IEEE safety requirements; coordinated with Engineering, Safety, Maintenance, and external contractors to mitigate potential operational losses exceeding \$1M
- Revised NFPA 70B- and IEEE-compliant electrical maintenance documentation for batteries, DC systems, UPS, and charger assets; led condition-based maintenance transition reviewed and approved across multiple facilities
- Developed CT wiring schematics and supporting scope-of-work for bus heater ammeter monitoring installation, from problem identification through contractor-ready documentation
- Selected for return internship (Summer 2026) based on performance and documented project impact

Take 5 Oil Change

Crosby, TX

Certified Lube Technician

March 2025 - May 2025

- Executed 40+ vehicle service cycles daily to manufacturer specifications in a fast-paced, safety-critical environment, maintaining procedural discipline and quality standards across high-volume operations

Planet Fitness

Crosby, TX

Member Services Representative

January 2024 - March 2025

- Managed operations for a 1,000+ member facility while converting an average of 6 prospects per week through structured presentations and relationship-driven communication

ORGANIZATIONS

Tau Beta Pi Honor Society

Member

Houston, TX

April 2025 - Present

Society of Hispanic Professional Engineers (SHPE)

Member

Houston, TX

August 2023 - Present

IEEE University of Houston Chapter

Member

Houston, TX

August 2025 - Present